Grasp-Conditioned Planning and Learning: Supplementary Material

Anonymous Author(s)

TABLE I
PER-ROBOT ACQUISITION REWARDS IN THE SEPARATE PPO COMPARISON. RATES ARE MULTIPLIED BY $\Delta t = 0.02\,\mathrm{s}$; EVENT REWARDS ARE ADDED ONCE. THE FROZEN FULL-SYSTEM ACTOR INSTEAD MINIMIZES IMITATION LOSS.

Term	Rate / event value
Approach progress	$6\,\mathrm{clip}((d_{t-1} - d_t)/\Delta t, -0.1, 0.1)$
Proximity	$2\exp[-(d_t/0.10)^2]$
Alignment, original	$\exp[-(\alpha_t/0.35)^2]$
Alignment, gated	$\exp[-(\alpha_t/0.35)^2 - (d_t/0.15)^2]$
Lift height	$10H(z_t - z^*)$
Lift levelness	$3\exp[-\lVert\eta_t\rVert^2/0.15^2]$
Approach / lift costs	$-0.5 - 4(\beta_i/\beta_{\max})^2$
Transition / failure	$+2$ each at both-attached and lift-complete; -5 on physical termination.

I. SCOPE AND ORGANIZATION

This supplement provides reproduction details and secondary experiments for *Grasp-Conditioned Planning and Learning for Multi-Robot Transport*. The frozen planar mission study, pickup PPO comparison, spatial probes and native-contact studies use distinct protocols. Their outcomes are not pooled.

II. REWARD SPECIFICATION

The full-mission actor minimizes the imitation objective in the weighted imitation objective in the main manuscript. This appendix specifies the separate pickup PPO and native-contact objectives; neither updates the frozen mission actor.

A. Pickup objective and optimization

Two separate PPO branches each run 65,536 steps with an asymmetric 181-input critic [1], [2]. They use generalized advantages [3], a 0.3 KL anchor, value coefficient 0.5 and zero entropy coefficient. Table I defines their acquisition rewards; the variants differ only in distance gating of orientation reward. Transport rewards retain common command-tracking, posture, grasp and regularization terms, detailed in Table II. All evaluations use deterministic actor means.

For maximum hand-position/orientation errors d_t, α_t, object roll/pitch η_t and base inclination β_i, the normalized height term is

$$\Phi(e) = \left(e^{-(e/0.08)^2} + 0.5e^{-(e/0.25)^2}\right)/1.5,$$
$$H(e) = \max\left\{0, \frac{\Phi(e) - \Phi(e_0)}{1 - \Phi(e_0)}\right\}, \tag{1}$$

where $e = z - z^*$ and $e_0 = z_{\mathrm{rest}} - z^*$. At zero initial error, $H = \Phi$. The inclination scale is $\beta_{\max} = 2°$.

Both branches use one environment, horizon 128 and 512 updates; the learning rate is 3×10^{-5}, with three epochs, four minibatches, clip ratio 0.2, $\gamma = 0.99$ and GAE $\lambda = 0.95$. Normalization is frozen and the critic receives 32 warmup updates. The network has ELU hidden layers 512–256–128. Mass is sampled from $U[0.75, 3]\,\mathrm{kg}$, dimensions from $U[0.9, 1.1]$, base offsets from 0.08–0.65 m, headings within $\pm 30°$, and activation delays within 0–2 s. COM perturbations are $\pm(0.05, 0.12, 0.05)\,\mathrm{m}$ and servo-gain factors 0.85–1.15. Automatic curricula are disabled.

Frozen pickup outcomes are 40/40 for the original reward and 39/40 for the distance-gated reward. Both stochastic training branches finish at 21/37 completed episodes. There is no demonstrated improvement from reward gating, and neither result establishes full-mission reliability.

B. Transport objective

For $K_\sigma(e) = \exp(-\lVert e\rVert^2/\sigma^2)$, the command-normalized kernel is

$$b = K_\sigma(c), \quad k = K_\sigma(x - c),$$
$$T_\sigma(x, c) = \frac{k - b}{\max(1 - b, 0.1)} + \left[1 - \min\left(\frac{1 - b}{0.1}, 1\right)\right] k. \tag{2}$$

Table II specifies every transport term in the archived PPO environment. Approach/lift replace this sum with Table I; they do not add both objectives.

C. Native-contact objective

The native-contact branch removes grasp welds and artificial attachment springs. Its per-robot objective combines progress, support and regularization with measured finger-contact costs:

$$r_{i,t} = \sum_{k \in \mathcal{K}_{\mathrm{task}}} w_k f_{k,i,t} + \sum_{k \in \mathcal{K}_{\mathrm{contact}}} w_k f_{k,i,t}. \tag{3}$$

Tables III and IV report guided-continuation (G) and the newer hand-target configuration (H) side by side; neither is retroactively assigned to earlier trials. Rates include $\Delta t = 0.02\,\mathrm{s}$; progress, event, work and effort increments do not receive a second timestep factor. Both actors receive local observations; training aggregates their trajectories centrally.

For hands $h \in \{L, R\}$, $\langle x_h \rangle = \frac{1}{2}\sum_h x_h$, d_h is palm-to-target distance, c_h closure, q_h raw contact qualification, a_h retained contact status, f_h firmness and m_h friction margin.

TABLE II

ARCHIVED PPO TRANSPORT OBJECTIVE. EVERY ROW IS MULTIPLIED BY $w_k \Delta t$; TERMINAL SEQUENCE FAILURE ADDS -5 ONCE. THESE REWARDS DO NOT OPTIMIZE THE FROZEN IMITATION ACTOR.

Term	**Weight**	**Raw contribution** f_k	**Definition / gate**
Command task	10	$e^{-\\|(\xi-\xi^*)/(0.15,0.15,0.2)\\|^2}$	Defines P; planar twist.
Linear tracking	3	$T_{0.25}(v, v^*)$	No gate.
Angular tracking	2	$T_{0.35}(\omega, \omega^*)$	No gate.
Height	2	$H(z - z^*)$	Multiplied by P.
Level	1	$K_{0.15}(\eta)$	Multiplied by P.
Grasp	1	$\mathbf{1}[\text{engaged}_i]$	Multiplied by P.
Base inclination	4	$-(\max_j \beta_j/(\pi/90))^2$	No gate.
Torso posture	0.5	$K_{0.261799}(\psi_i)$	Multiplied by P.
Arm symmetry	0.5	$K_{0.15}(e_{\text{sym,i}})$	Multiplied by P; metric hand-symmetry error.
Contact constellation	1	$-E_{c,i}/0.05^2$	Mean squared palm-landmark error.
Base constellation	1	$-\mathbf{1}[\text{engaged}_i]E_{b,i}/0.05^2$	Mean squared base-landmark error.
Residual magnitude	2	$-\\|a_{0:3}\\|^2 - 0.1\\|a_{3:}\\|^2$	Base and upper-body action costs.
Object tilt	0.5	$-\min((\\|\eta\\|_\infty/0.15)^2, 4)$	No gate.
Action change	0.005	$-\\|a_t - a_{t-1}\\|^2$	No gate.
Drop	50	$-\mathbf{1}[z < z_{\text{rest}} - 0.12]$	No gate.

TABLE III

COMPLETE NATIVE-CONTACT TASK OBJECTIVE. WEIGHTS G/H DENOTE GUIDED CONTINUATION / THE NEWER HAND-TARGET CONFIGURATION. ZERO DISABLES A TERM; TIMESTEP AND EVENT CONVENTIONS ARE EXPLICIT.

Term	**G**	**H**	**Raw contribution** f_k	**Definition**
Grasp progress	20	20	$\text{clip}(g_{t-1} - g_t, -.05, .05)$	Metric grasp-error reduction.
Rotation progress	0	6	$\text{clip}(\alpha_{t-1} - \alpha_t, -.02, .02)$	Angular grasp-error reduction.
Contact potential	5	5	$\Phi_{i,t} - \Phi_{i,t-1}$	Distance, closure and retained contact.
Grasp loss	6	6	$-N_{\text{release}}$	Excludes injected self-opening.
Proximity	1	1	$\Delta t e^{-(g/.08)^2}$	Dense approach reward.
Supported progress	100	100	$G(\Delta s)$	Path progress.
Goal progress	150	200	$G(\\|De_{t-1}\\| - \\|De_t\\|)$	Translation and rotation.
Supported lift progress	0	200	$G(\|e_{z,t-1}\| - \|e_{z,t}\|)$	Supported height-error reduction.
Stable goal	2	2	$\Delta t \mathbf{1}[\text{good}]$	Supported stable-goal condition.
Unsupported travel	20	20	$-\Delta \ell_{xy} \mathbf{1}[\neg\text{supported}]$	Horizontal object displacement.
Anchor slip	2	2	$-\Delta t \min(\delta, .1)\mathbf{1}[\text{engaged}_i]$	Captured-frame grasp drift.
Load	0.2	0.2	$-\Delta t(\\|F_i\\|/\max(1, 2F_{\text{cap}}))^2$	Configured load normalization.
Object orientation	4	4	$-\Delta t \\|e_R\\|^2$	Nearby-reference rotation error.
Torso posture	0.2	0.2	$-\Delta t \psi_i^2$	Local torso pitch.
Action change	0.01	0.01	$-\Delta t \\|a_t - a_{t-1}\\|^2$	Normalized actor outputs.
Time	2	2	$-\Delta t$	Per-second cost.
Success	300	300	$\mathbf{1}[\text{success}]$	Terminal bonus.
Failure	500	500	$-\mathbf{1}[\text{terminated}]$	Physical termination penalty.

The acquisition potential and support gate are

$$\Phi_i = \sum_h \left[e^{-(d_h/0.04)^2} \text{clip}(c_h/c^*, 0, 1) + a_h \right],$$

$$G(v) = \begin{cases} v, & \text{supported at both interval endpoints,} \\ \min(v, 0), & \text{otherwise.} \end{cases} \tag{4}$$

The position–rotation error uses $D = \text{diag}(1, 1, 1, \rho, \rho, \rho)$, $\rho = 0.35\,\text{m}$. Native contact shaping uses $\sigma_d = 0.06\,\text{m}$, $\sigma_v = 0.05\,\text{m/s}$ and $F_{\max} = 40\,\text{N}$; the separate load normalization uses $F_{\text{cap}} = 60\,\text{N}$. Finger work is $\Delta W_i = \int \sum_j |\tau_j \dot{q}_j| dt$ and effort is $\Delta E_i = \int \sum_j \tau_j^2 dt$. The contact-frame cost uses four landmarks $\{0, 0.04e_x, 0.04e_y, 0.04e_z\}$ per palm relative to the assigned or captured object frame. Recorded contact proxies are simulator diagnostics, not hardware-calibrated force-closure certificates.

For contact normals n_a on distinct finger bodies, the opposition score, normal-force total and force-weighted slip are

$$\begin{aligned} o_h &= \max_{a,b} \frac{1 - n_a^\top n_b}{2}, \quad F_{n,h} = \sum_a F_{n,a}, \\ v_{\text{slip,h}} &= \frac{\sum_a F_{n,a} v_{\text{slip,a}}}{\max(F_{n,h}, 10^{-12})}, \\ f_h &= \min(F_{n,h}/12, 1) o_h, \\ m_h &= \text{clip}(1 - \max_a \chi_a, 0, 1), \\ q_h &= \mathbf{1}[o_h \geq 0.7,\ F_{n,h} \geq 2,\ v_{\text{slip,h}} \leq 0.04], \end{aligned} \tag{5}$$

where χ_a is contact friction utilization. With no contact, all quality quantities are zero. First acquisition requires $0.08\,\text{s}$ qualification; retained status tolerates at most $0.02\,\text{s}$ unqualified contact, unless normal force falls below $0.2\,\text{N}$. These temporal filters affect observation and reward bookkeeping, not attachment forces.

The hand-target extension adds pose alignment, measured closure progress and relative held-hand motion, together with

TABLE IV

COMPLETE NATIVE-CONTACT CONTACT OBJECTIVE. WEIGHTS G/H DENOTE GUIDED CONTINUATION / THE NEWER HAND-TARGET CONFIGURATION. ZERO DISABLES A TERM; TIMESTEP AND EVENT CONVENTIONS ARE EXPLICIT.

Term	**G**	**H**	**Raw contribution f_k**	**Definition**
First acquisition	8	8	ΔN_i	One event per hand's first qualified acquisition.
Distance	6	6	$-\Delta t\langle\tanh(d_h/\sigma_d)\rangle$	Metric palm-target separation.
Separation speed	3	3	$-\Delta t\langle\mathrm{clip}(v_{\mathrm{out,h}}/\sigma_v, 0, 1)\rangle$	Outward relative velocity.
Secure contact	4	0	$-\Delta t\langle 1-q_h\rangle$	Qualified opposing contacts.
Firmness	1.5	1.5	$-\Delta t\langle 1-f_h\rangle$	Contact-quality proxy.
Contact slip	12	12	$-\Delta t\langle v_{\mathrm{slip,h}}\rangle$	Tangential slip speed.
Friction reserve	0.5	0.5	$-\Delta t\langle 1-f_h m_h\rangle$	Firmness-weighted friction margin.
Excess force	2	2	$-\Delta t\langle\max(F_{n,h}/F_{\max}-1,0)^2\rangle$	Normal-force overage.
Finger work	0.02	0.02	$-\Delta W_i$	Absolute actuator work increment.
Holding effort	0.002	0.002	$-\Delta E_i$	Squared-torque integral.
Premature closure	0.5	0.5	$-\Delta t\sum_h \mathbf{1}[d_{\mathrm{nom,h}}>.06]c_h^2$	Closure away from the nominal target.
Contact frame	100	100	$-\Delta t\sum_h \mathrm{mean}_k\,\Vert p_h+R_hu_k-p_h^*-R_h^*u_k\Vert^2$	Position and orientation via four landmarks.
Pose alignment	0	3	$-\Delta t\langle 1-A_h\rangle$	Assigned or captured palm frame.
Aligned closure	0	12	$\Psi_t-\Psi_{t-1}$	Measured finger flexion, weighted by alignment.
Held-hand adjustment	0	1	$-\Delta t\langle a_hJ_h\rangle$	Motion relative to the rigid object at the hand.

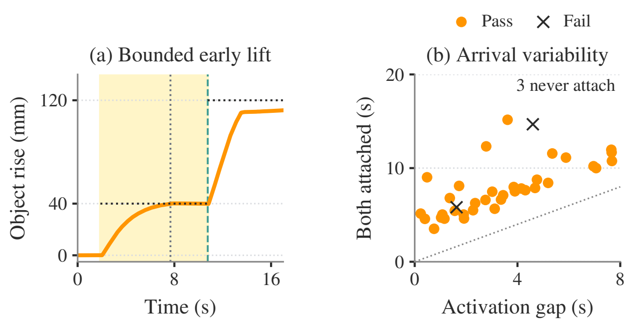

Fig. 1. Independent arrival. (a) Loading dock, seed 142013: measured object rise under a 7.68 s activation gap. Shading marks early lift; vertical dotted/dashed lines mark late activation/second attachment; horizontal lines mark height targets. (b) Both-attachment times for the 37 trials that acquire the object; color marks final mission outcome. The other three trials never attach both robots. The diagonal is $t=\Delta$.

supported lift progress:

$$A_h = e^{-(d_h/.03)^2-(\alpha_h/.25)^2},$$
$$\Psi_i = \langle A_h\,\mathrm{clip}(\bar{c}_h/.6,0,1)\rangle. \tag{6}$$

Here $\bar{c}_h$ is mean normalized MCP/PIP flexion of the non-thumb fingers, measured from joint encoders. Relative velocities subtract the object's rigid motion at the actual hand position. Tables III–IV specify both configurations independently.

$$J_h = \mathrm{clip}\left(\frac{\Vert v_{\mathrm{rel,h}}\Vert^2}{.03^2}+\frac{\Vert\omega_{\mathrm{rel,h}}\Vert^2}{.2^2},0,1\right). \tag{7}$$

III. Asynchronous Acquisition Trace

IV. Planar Imitation Protocol

The mission actor is trained by imitation and DAgger [4]. A geometric teacher labels approach/lift, and a velocity actor labels transport. Three aggregation rounds collect 30 episodes each: teacher execution first, then student execution with teacher labels and no interventions. Stable-hold collection succeeds in 86/90 episodes. The final dataset contains 524,490 local observations, including 345,484 imported samples. Phase-balanced weighted action MSE uses the settings in Table V and frozen normalization. The mission actor receives no subsequent PPO update.

Imported data comprise 227,902 samples from the earlier all-object actor and 117,582 from the independent-arrival actor, totaling 65.9% of the final dataset. In 22,205 early-lift rows, the height-reference feature was rebased from 120 to 40 mm; historical action labels, other features and observation noise were retained. These are reused development data, not an independent validation set. The initialization and dataset hashes are retained with the supplementary provenance record.

The transport teacher is the saved shape-conditioned velocity actor `velocity_v6_shapes`. In furnished scenes, collection commands zero transport velocity after lifting; navigation is supplied by RRT/MPC at test time. Thus successful collection is not evidence of learning complete doorway missions from demonstrations. The sampler balances the nonempty approach, lift and transport groups; the implementation minimizes

$$\mathcal{L} = \frac{1}{|B|}\sum_{j\in B}\frac{1}{21}\sum_{k=1}^{21} w_k\big(\pi_\theta(\bar{o}_j)_k - a^*_{jk}\big)^2. \tag{8}$$

Here $\bar{o}$ uses frozen observation normalization. The bootstrap script's recorded defaults are batch size 512 and weights $w=(4,4,2,4,0.5,1,\ldots,1)$; equal phase allocation yields 510 samples when all three groups are present. The saved run metadata do not independently record every fitting CLI override, limiting exact reconstruction from metadata alone. Configuration and implementation details accompany the manuscript.

DAgger collection randomizes mass $U[1,2]$ kg, dimension factors $U[0.95,1.05]$, rail friction $U[0.9,1.8]$, base offsets 0.08–0.50 m, headings $\pm 20°$ and activation delays 0–8 s across ten scenes. Automatic stage/difficulty curricula are disabled; DAgger changes the executing teacher/student across rounds. Training ends at 35 s by truncation or earlier upon physical failure; successful DAgger collections also stop early. Figure 2 reports optimization and stochastic collection outcomes, distinct from the frozen evaluations.

TABLE V

COMPACT LEARNING AND EXECUTION SETTINGS.

Component	Configuration
Actor input / output	157 observations; 21 actions per robot.
Network	ELU MLP, 512–256–128 hidden units.
Physics / policy	200 / 50 Hz; identical actor weights.
Imitation	Adam 3×10^{-4}; 3 fits, 40 epochs each.

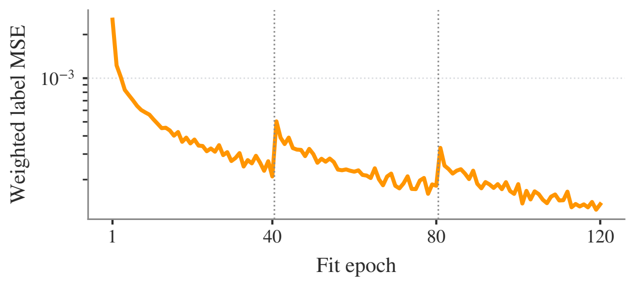

Fig. 2. Imitation fitting across three 40-epoch rounds; dotted lines mark dataset aggregation. Collection succeeds in 30/30 teacher episodes, then 28/30 and 28/30 student episodes. This is training evidence, not a matched comparison with the teacher at full-mission evaluation.

V. SPATIAL REFERENCE AND PROBE PROVENANCE

The spatial imitation actor uses 54 collection episodes in three rounds, 60 fitting epochs per round, and planner-selected targets from $\{0^\circ, 15^\circ, 45^\circ\}$. Initialization transfers the upper/lower-grasp planar actor. After the first round, the configured teacher fraction is 0.25. The gallery-panel mass is fixed at 1.5 kg; activation is synchronized, without asynchronous early lift. Pose observations append local position/rotation error, desired and measured spatial twist, contact geometry and stance error. The actor still outputs 21 commands, with bounded arm/lift rates during spatial tracking. Pose references and phase information remain shared in training and execution. The 90° demonstration is geometric-teacher control, not this trained actor. These probes receive no PPO update; subsequent spatial PPO runs are separate development studies.

The illustrated teacher probes use upper/lower rail spacing 0.20 m; the learned probes use 0.16 m. A 0.15 m clearance lift takes 8 s after a 2 s post-lift hold. Angular interpolation takes 22.5 s for 45° and 45 s for 90°. The generic historical reference generator also permits 180° commands; the manuscript's 90° fixed-grasp envelope is a declared scope restriction, not a retrospectively applied termination guard. Roll is relative to the initial lifted object frame. No large-roll probe establishes doorway passage or all-pose reachability.

The complete same-start learned test contains 0°, 15° and 45° targets: two of three pass. The five website illustrations above are selected probes. The adaptive teacher index retains all 42 attempts (six passing), with changing geometry and controller settings; neither that count nor the selected successes estimates frozen-controller reliability. The failed 180° trial stops before 90° and therefore cannot identify 90° as an experimentally measured failure threshold.

TABLE VI

SPATIAL PROBES. EACH ROW IS ONE NOMINAL TRIAL; CONFIGURATIONS DIFFER.

Controller	Goal	Final roll	Outcome
Teacher	45°	44.29°	Pass
Teacher	90°	88.29°	Pass
Teacher (outside scope)	180°	51.40°	Tracking failure
Learned actor	15°	12.83°	Pass
Learned actor	45°	17.26°	Self-penetration

TABLE VII

LATER FROZEN PLANAR COHORT: ALL TWELVE SCENES.

Scene	Completed / declared
Apartment	4/4
Bedroom	4/4
Open arena	4/4
Gallery panel	4/4
Living room	4/4
Living-room painting	3/4
Loading dock	4/4
Office table	4/4
Dining room	4/4
Warehouse crate	4/4
Warehouse panel	3/4
Workshop beam	4/4
Overall	46/48

VI. LATER PLANAR COHORT

The twelve-scene evaluation fixes the actor checkpoint and declares 48 starts before execution. It uses 1–2 kg loads, $\pm5\%$ dimensions, 0.08–0.50 m base offsets, $\pm20^\circ$ headings, 0.10 m object-position jitter and up to 8 s activation delay. The budget is 180 s and planned clearance is 20 mm. Upper/lower handles extend panel acquisition; supervised progression and idealized weld retention remain. Its different actor and source configuration prevent attribution of the difference from 35/40 to a single modification. The two acquisition timeouts occur in the living-room painting and warehouse panel scenes. Every declared trial remains in the denominator. This cohort has no large-roll task and does not replace the original formation/clearance analysis. Actor hash, source manifest and per-scene outcomes are retained in `spatial_evidence.json`.

VII. LOCAL-HISTORY COORDINATION: IMPLEMENTED DEVELOPMENT TRIALS

A separate controller uses a shared causal transformer with private per-robot history, breakable finite attachments and local measurement ages. Its actor omits partner state, coordination phase and injected-fault flags. The spatial reference follows object progress; this differs from the timed spatial probes. A compact statement of the training/execution split is

$$a_{i,t} = \pi_\theta(o_{i,t-H+1:t}), \qquad V_\psi(s_t^{\text{priv}}), \\ s_t^{\text{priv}} = [q_t, 0.1\dot{q}_t, b_{1,t}, b_{2,t}, f_t]. \tag{9}$$

where $q_t, \dot{q}_t$ contain both robots and the object, b_i are attachment indicators and f_t indicates fault injection. The centralized critic is used only in PPO training. Imitation

TABLE VIII

COMPLETED LOCAL-HISTORY DEVELOPMENT SUITES. COLUMNS ARE DISTINCT TASKS, NOT A POOLED SCORE.

Variant	Transport	Wait	Early release	Motion
V44	12/24	4/8	6/8	14/14
V45	12/24	7/8	7/8	12/14
V46	12/24	3/8	7/8	13/14
V47	0/24	0/8	0/8	0/14

centrally pools teacher-labeled robot samples; PPO pools rollouts and returns. Neither labels, partner histories, critic inputs nor optimizer state are supplied to the executing actors. Shared task references and simulated object estimates are still information inputs; no network budget is measured.

These four consecutive completed development variants are reported together rather than selecting the best score. V44 uses mixed-state imitation; V45 adds a current-observation head; V46 applies PPO; V47 varies release progress during collection. They are different training variants, not independent training seeds or a communication ablation.

The full suite contains 24 transport tasks and eight absent-partner waits. Transport comprises twelve nominal/delayed starts, eight later releases and four stale-pose cases. Later releases require four-hand support, object rise above 30 mm and path progress above 150 mm. Early-release and signed-motion suites have separate denominators. Safe waiting is not counted as transport completion. All variants reject payload/non-hand robot contact above 1 N; unsupported travel remains limited to 50 mm. V45 and V46 solve none of the eight later-release or four stale-pose cases. V47 fails all suites despite successful mixed-control collection, illustrating the gap between supervised fitting and closed-loop coordination. These results do not establish robust local recovery, broad scene transfer or learned 90° reorientation.

VIII. FROZEN PLANAR FEEDBACK SEMANTICS

The frozen actor uses the legacy constraint feature: the first six equality solver multipliers for each palm weld are summed by row index, scaled by 0.01 and clipped to $[-5, 5]$ before observation normalization. The first three rows constrain translation and the next three orientation. This concatenation is not transformed into a calibrated robot-frame spatial wrench; in particular, the rotational rows cannot be interpreted directly as wrist torques. It is a simulator-specific feature, not validated force sensing. Removing or replacing it requires evaluation with the appropriate observation distribution.

Contact gates test acquisition, but active welds can sustain tensile forces and moments without a frictional retention limit. Recorded actuator models do not, by themselves, validate hand force capacity. The study reports neither actuator saturation distributions nor a load-to-capacity ratio; the 1–2 kg objects test large geometry rather than heavy-payload capability. Frictional slip and hardware sensor equivalence remain untested. The separate local-history suites inject loss of support under finite attachments; those results do not validate the original welded-retention model.

TABLE IX

EVALUATION PROTOCOLS AND ORIGINAL DENOMINATORS.

Test	Domain / comparison	Pass
Frozen missions	10 scenes; 1–2 kg; independent starts, $\Delta \leq 8$ s; 180 s.	35/40
Original MPC	8 scenes; 1.5 kg; gated starts, $\Delta \leq 0.8$ s; 180 s.	36/40
Aligned MPC	Same actor and 40 seeds as original MPC; geometry and margin change.	40/40

IX. ADDITIONAL SPATIAL DIAGNOSTICS

X. GEOMETRIC ILLUSTRATION

XI. ADDITIONAL PLANAR DIAGNOSTICS

XII. NATIVE-CONTACT ACQUISITION: SCOPE AND OUTCOMES

Native-contact experiments remove both grasp equalities and artificial hand–object springs. Load transmission instead depends on articulated finger contacts, finite actuator limits and randomized friction. Their objective terms appear in Tables III and IV. These experiments are separate from the welded mission evaluation and do not establish native-contact transport.

An actuator audit identified missing inheritance of hand motor defaults in the initial implementation. The corrected model sets finger/wrist position gains to 2 N m/rad and torque limits to ± 1 N m. A scripted four-hand acquisition, 6 cm lift and 8 s hold succeeds without attachment forces. Paired 500/1000 Hz simulations differ by at most 1.25 mm and 0.16° in object trajectory. This tests a short scripted trajectory, not learned robustness. Pre-correction failures remain in the archive and are excluded from corrected-model acceptance.

A paired 32,768-step PPO study uses identical initial weights, training seed and 20 evaluation seeds. The distance/outward-speed variant attains four-hand support in 18/20 trials but completes 0/20 lift-and-hold tasks: 17 fail on unintended payload–robot contact, two on self-contact and one on unsupported travel. The missing-contact-cost variant achieves neither acquisition nor completion (0/20); the baseline acquires in 2/20 and completes 0/20. These results distinguish contact acquisition from sustained task success; one training seed does not establish a general reward ranking.

Teacher initialization similarly fails to transfer reliably: collections with 10/12, 20/20 and 24/24 successful teacher trials each yield 0/20 fresh frozen-actor successes in their respective configurations. The corrected cross-scene screen completes 0/26 cases. Later corrective-label, architecture and reward experiments are development data and do not update these frozen cohorts. A subsequent calibrated near-grasp lift policy is reported in the expanded evidence section below. Full-mission native-contact reliability remains unestablished.

XIII. PLATFORM AND PROSPECTIVE MOTOR INTERFACE

Each carrier has a holonomic three-module steering/drive base, a two-stage lift, a pitching torso, two 7-DoF arms and two hands with 16 finger DoFs and an additional wrist each (main manuscript architecture). The planar policy commands three base and eighteen upper-body coordinates;

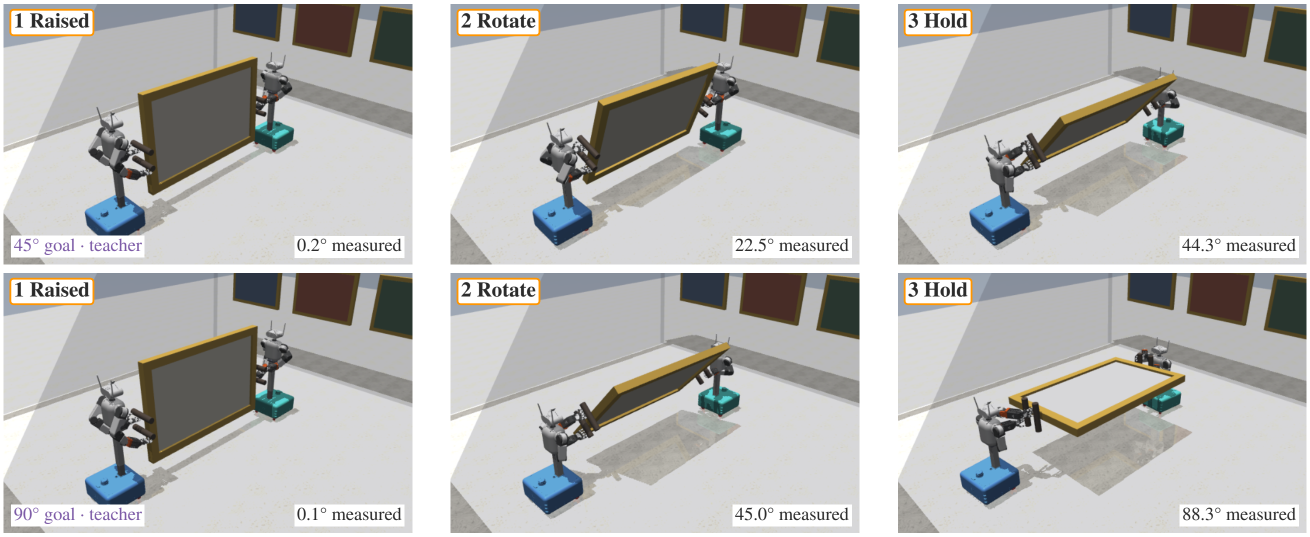

Fig. 3. Recorded fixed-grasp panel reorientation with the geometric teacher (seed 207000). Rows request 45° and 90° roll about the lifted object's long axis. Columns show clearance lift, intermediate rotation and terminal holding; labels give measured angles. Robot A/B have blue/teal bases and native body materials. Original joint states are rendered without stepping dynamics. These are pose-only probes, not learned-policy or doorway-transport successes.

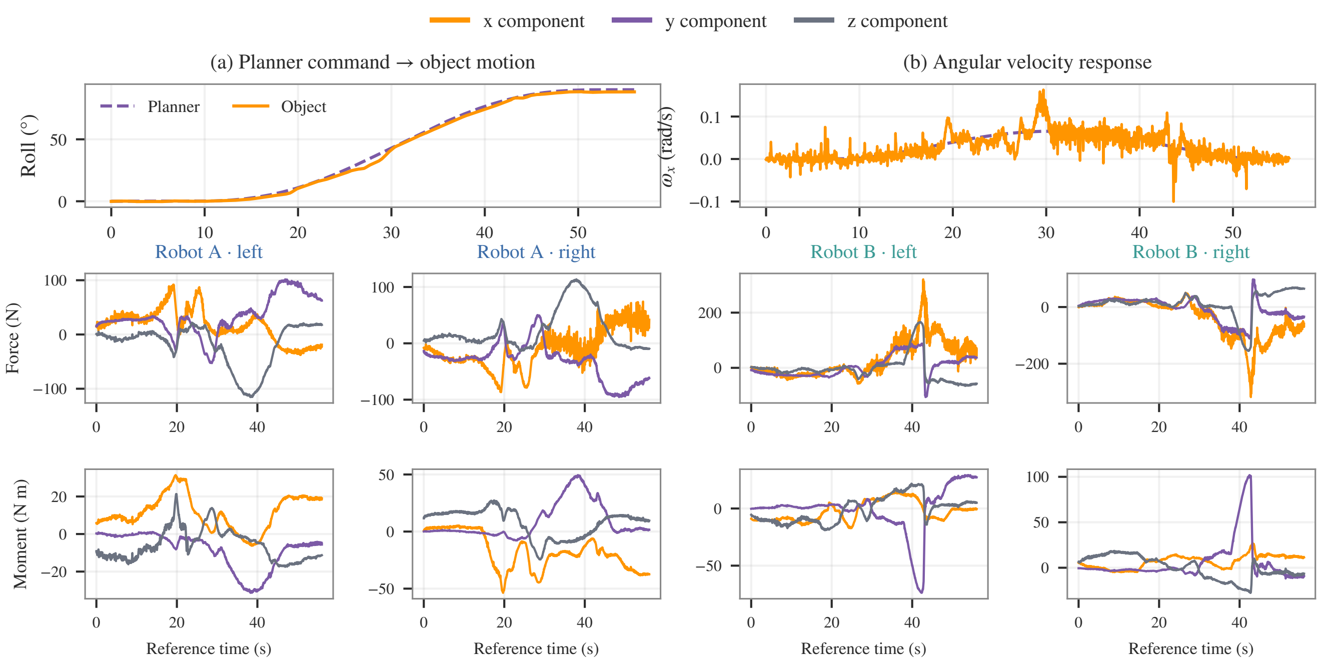

Fig. 4. Commanded object motion and resulting four-hand weld reactions in one instrumented 90° teacher probe. Dashed purple: pose/twist reference; orange: measured object response. Hand components use fixed lifted-object axes (x: orange, y: purple, z: gray); moments are palm-centered. Time starts at pose-schedule activation; ordinate scales vary by panel. These are simulated reactions, not commanded force targets or wrist-sensor data.

hand articulation uses stage-dependent closure rather than learned finger-level actions. The hardware interface uses Jetson Thor for local inference and CAN arm-drive commands $(q^d, \dot{q}^d, K_p, K_d, \tau^{\mathrm{ff}})$:

$$\tau^d = \mathrm{clip}\big(K_p(q^d - q) + K_d(\dot{q}^d - \dot{q}) + \tau^{\mathrm{ff}},\ -\tau_{\max}, \tau_{\max}\big)\,. \tag{10}$$

Embedded field-oriented control (FOC) regulates arm-motor current; MuJoCo uses its actuator model instead. The 4 kg hardware target load is distinct from the evaluated 1–2 kg simulation loads. Hardware system performance and that load target are not validated by the reported experiments.

XIV. Reconstruction of Individual Hand Reactions

For hand h of robot i, selected weld multipliers λ_{ih} give the payload wrench through

$$J_O^\top w_{ih}^{O,W} = S_O J_{ih}^\top \lambda_{ih},$$

$$w_{ih} = \begin{bmatrix} R_0^\top F_{ih}^W \\ R_0^\top [\tau_{ih}^{O,W} - (p_{ih}^W - p_O^W) \times F_{ih}^W] \end{bmatrix}. \tag{11}$$

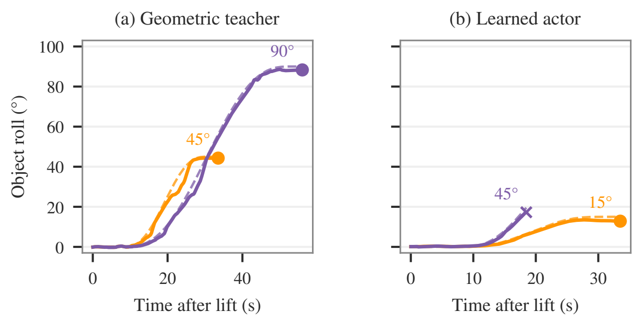

Fig. 5. Measured roll (solid) and evolving reference (dashed), relative to the lifted object frame. Circles mark successful completion; the cross marks self-penetration termination. Labels specify commanded endpoints, which are distinct from the evolving reference. Teacher and actor configurations differ.

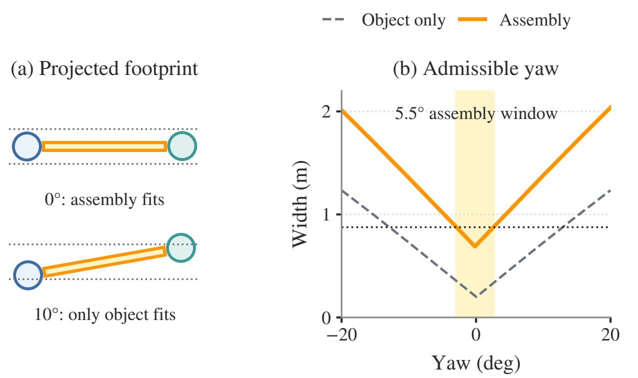

Fig. 6. Why the grasp changes passage feasibility. Analytic projections use the measured beam handoff geometry (seed 142024). (a) Dashed bounds show the available opening width after the 22 mm search margin on each side; the circles are base footprints. (b) The object-only yaw range is about $25.8°$, versus $5.5°$ for the assembly (shaded); the horizontal dotted line is the door budget.

Here S_O selects the free payload's six generalized coordinates, J_O maps their velocities to its world twist, and J_{ih} is the weld Jacobian. R_0 fixes the lifted-object axes; moments act about the measured palm center p_{ih}. These solver reactions act on the payload and exclude separate finger-contact forces.

XV. Expanded Geometric and Native-Contact Evidence

A. Grasp-space studies and matched search

The perimeter and end-face studies are separate enumerations. The former contains 2,110,211 opposed candidate layouts, 45,917 geometrically feasible layouts and 51 verified routes among 54 selected layouts. The latter checks 108 selected layouts with whole-body IK (95 feasible) and 54 selected routes (52 verified). These are discrete 50 mm screens and selected downstream checks, not coverage guarantees over continuous grasp assignments.

The new matched sampling study uses apartment-scene seeds 0–15, a 20,000-iteration budget, 0.35 m steering, goal bias 0.10, margin 0.02 m, 10 mm swept checking, explicit tree-join validation, no portal seeding and no smoothing. Gate probability alone changes, between 0.35 and zero. All returned

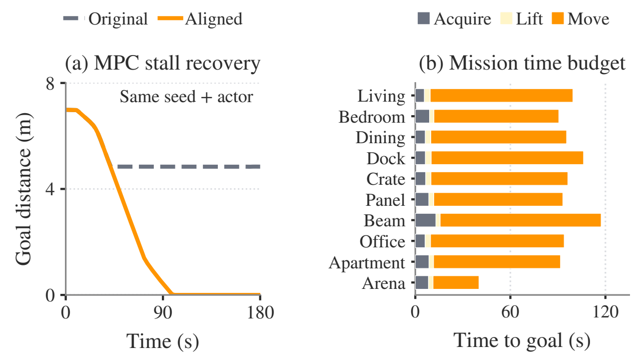

Fig. 7. Planning consistency and mission duration. (a) Same-seed, same-actor MPC comparison (114000): the original model stalls before the doorway; aligned geometry and margin permit progress to the goal. (b) Mean phase durations among successful frozen missions in each scene. Navigation dominates elapsed simulation time. The paired development case and frozen test use different actors and domains.

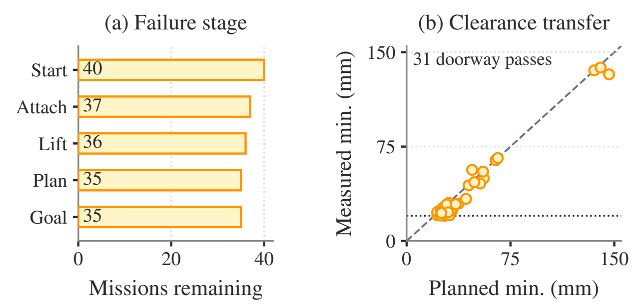

Fig. 8. (a) Mission counts at each stage, retaining all 40 starts. (b) Planned versus measured minimum clearance for 31 successful doorway missions; four arena passes have no finite clearance. Diagonal: equality; horizontal dotted line: 20 mm planning requirement.

routes are independently checked at 5 mm swept resolution. Verified counts are 16/16 and 5/16. Timing is recorded but not used to claim real-time capability; the machine was concurrently running other workloads. The legacy notebook's eight-seed counts are not substituted for this study.

B. Native checkpoint and task identity

The native lift checkpoint has SHA-256

```
e2ec616a001dbb03cea71ee68a434c090199
eecef0e88e3246ad2c7d117b1ae7
```

and 81,920 environment steps (80 updates). Its frozen configuration contains an eight-frame, 2,816-input, two-layer 128–128 ELU actor, a 282-input critic and 23 actions per robot. PPO uses eight environments, horizon 128, four optimization epochs, four minibatches, $\gamma = 0.995$, GAE $\lambda = 0.95$, clip ratio 0.2, initial learning rate 10^{-4}, entropy coefficient 0.0005 and frozen input normalization. Actor weights are not initialized from an imitation checkpoint. The two gripper output biases begin at $+1$ and remain trainable; initial log standard deviation is -0.8.

The base translation cap is 0.001 m/s and the base angular action scale is 0.001 rad/s; these are small nonzero authorities, not mechanical locks. Arm and torso position scales are 0.08 and 0.05 rad. The nominal payload is 0.3 kg, with calibrated palm offsets $(-65, +70)$ mm and no distant approach at reset. Commanded height varies over 0.06–0.14 m. The independent hold-pose acceptance requires one continuous second with object position and orientation error below 0.04 m and 0.08 rad, every palm below 0.04 m and 0.175 rad error, every grasp qualified, and object speeds below 0.05 m/s and 0.2 rad/s. This differs from the older full-approach 8 s-hold evaluations; their rates are not pooled or treated as a matched improvement.

C. Evaluation-domain correction

The 200-trial report uses seeds 7,500,000–7,500,199 and the fixed checkpoint above. All 200 trials satisfy the hold-pose criterion. Its original `difficulty=1.0` metadata predates the evaluator correction that writes this value into the environment's randomization scale. Setting the training scheduler's start value while disabling that scheduler did not change the environment. The manuscript therefore labels this cohort by the nominal randomization level and makes no full-domain claim from it. The original report is preserved byte-for-byte, with its interpretation in the evidence manifest. Subsequent full-domain evaluations and retraining are outside this frozen manuscript snapshot.

The evaluator also emits `time_limit` when a successful episode has no failure-cause string. This placeholder is not treated as evidence that all trials ran to their time budget. Reported completion times come from the measured first-success timestamp (median 5.07 s), and the 95% Wilson interval 98.1–100% is conditional on this task, checkpoint and seed domain. The training plot uses a rolling 20-completed-episode mean; no uncertainty band is inferred from a single training run.

D. Reproducing the figures

The directory `data/ral/expanded` contains the selected original records, checkpoint configuration, source hashes, individual search trials and plot summaries. The original run was subsequently archived under `grasp_carry_L_lift_v1_weakDR`; its identity is the checksum above, not the reusable run-directory name. The figure script reads the frozen copies, so ongoing training cannot change the publication plots. The authored training configuration preserves base reward weights; the checkpoint snapshot can contain weights already rescaled by the curriculum. `make journal-figures` regenerates the three vector figures without robot simulation. The sampling benchmark is rerun separately through `scripts/benchmark_gate_sampling.py` in an M2 checkout.

XVI. Reproduction record

The accompanying `data/ral` records retain source hashes, imported-data provenance, reward definitions, actor/configuration identities and original outcomes. A review-response matrix distinguishes completed editorial and post-hoc analyses from missing controlled experiments. The spatial supplement freezes the site-linked probe sources and completed V44–V47 recovery suites; ongoing trials are excluded. The full-mission actor uses imitation; the reward table above is specific to this separate PPO study.

References

[1] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv:1707.06347*, 2017. [Online]. Available: https://arxiv.org/abs/1707.06347

[2] L. Pinto, M. Andrychowicz, P. Welinder, W. Zaremba, and P. Abbeel, "Asymmetric actor critic for image-based robot learning," *arXiv:1710.06542*, 2017. [Online]. Available: https://arxiv.org/abs/1710.06542

[3] J. Schulman, P. Moritz, S. Levine, M. Jordan, and P. Abbeel, "High-dimensional continuous control using generalized advantage estimation," *arXiv:1506.02438*, 2015. [Online]. Available: https://arxiv.org/abs/1506.02438

[4] S. Ross, G. Gordon, and D. Bagnell, "A reduction of imitation learning and structured prediction to no-regret online learning," in *Proc. Int. Conf. Artif. Intell. Stat. (AISTATS)*, ser. Proceedings of Machine Learning Research, vol. 15, 2011, pp. 627–635. [Online]. Available: https://proceedings.mlr.press/v15/ross11a.html